

4. Poisson Distribution — Solved Problems in Data Science

1. What is Poisson Distribution?

The **Poisson distribution** is used to model the number of occurrences of an event in a fixed interval when:

- Events occur independently.
- The average rate of occurrence is approximately constant.
- Two events do not occur at exactly the same instant in the idealized model.
- We are interested in the **count of events**.

Examples in Data Science:

- Number of website visitors per minute
- Number of customer complaints per day
- Number of server requests per second
- Number of fraudulent transactions per hour
- Number of errors in a dataset
- Number of machine failures per month
- Number of clicks received per minute

If:

$$X \sim \text{Poisson}(\lambda)$$

then:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

where:

- X = number of events
- x = specific number of events
- λ = average number of events in the interval
- $e \approx 2.71828$

2. Important Properties

For a Poisson distribution:

Mean

$$E(X) = \lambda$$

Variance

$$\text{Var}(X) = \lambda$$

Standard deviation

$$\sigma = \sqrt{\lambda}$$

An important property is:

$$\text{Mean} = \text{Variance} = \lambda$$

This is one of the easiest ways to recognize a Poisson distribution.

Problem 1: Website Visitors

A website receives an average of **5 visitors per minute**.

Find the probability that exactly **3 visitors** arrive in a particular minute.

Solution

Given:

$$\lambda = 5$$

We need:

$$P(X = 3)$$

Using the Poisson formula:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Therefore:

$$P(X = 3) = \frac{e^{-5} 5^3}{3!}$$

Calculate:

$$\begin{aligned}5^3 &= 125 \\ 3! &= 6\end{aligned}$$

Therefore:

$$P(X = 3) = \frac{e^{-5}(125)}{6}$$

Since:

$$e^{-5} \approx 0.0067379$$

we get:

$$P(X = 3) \approx 0.1404$$

Answer

$$P(X = 3) \approx 0.1404$$

or

$$14.04\%$$

Data Science Interpretation

There is approximately a **14.04% probability that exactly 3 visitors arrive during a particular minute.**

This can be useful for **website traffic analysis and server capacity planning.**

Problem 2: Server Requests

A web server receives an average of **10 requests per second.**

What is the probability that the server receives exactly **8 requests in one second?**

Solution

Given:

$$\begin{aligned}\lambda &= 10 \\ x &= 8\end{aligned}$$

Using:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Therefore:

$$P(X = 8) = \frac{e^{-10} 10^8}{8!}$$

Since:

$$8! = 40320$$

we obtain:

$$P(X = 8) = \frac{e^{-10} (100000000)}{40320}$$

$$P(X = 8) \approx 0.1126$$

Answer

$$11.26\%$$

Interpretation

There is approximately an **11.26% probability of receiving exactly 8 requests in one second.**

Problem 3: Customer Complaints

A company receives an average of **4 customer complaints per day.**

Find the probability of receiving exactly **2 complaints on a particular day.**

Solution

Given:

$$\begin{aligned}\lambda &= 4 \\ x &= 2\end{aligned}$$

Therefore:

$$P(X = 2) = \frac{e^{-4}4^2}{2!}$$

$$P(X = 2) = \frac{e^{-4}(16)}{2}$$

$$P(X = 2) = 8e^{-4}$$

Since:

$$e^{-4} \approx 0.0183156$$

we get:

$$P(X = 2) \approx 8(0.0183156)$$

$$P(X = 2) \approx 0.1465$$

Answer

$$14.65\%$$

Problem 4: Fraudulent Transactions

A banking system observes an average of **2 fraudulent transactions per day**.

Find the probability that exactly **3 fraudulent transactions** occur on a particular day.

Solution

Given:

$$\lambda = 2$$

$$x = 3$$

Using:

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

we get:

$$P(X = 3) = \frac{e^{-2}2^3}{3!}$$

$$P(X = 3) = \frac{e^{-2}(8)}{6}$$

$$P(X = 3) = \frac{4}{3}e^{-2}$$

Since:

$$e^{-2} \approx 0.135335$$
$$P(X = 3) \approx 0.1804$$

Answer

$$P(X = 3) \approx 18.04\%$$

Data Science Application

This can help a fraud analytics team estimate the probability of seeing a particular number of suspicious transactions in a day.

Problem 5: Machine Learning Errors

An ML model produces an average of **3 prediction errors per 1,000 predictions**.

Find the probability of exactly **2 errors in the next 1,000 predictions**.

Solution

Here the observation interval is:

1000 predictions

The average number of errors is:

$$\lambda = 3$$

We need:

$$P(X = 2)$$

Therefore:

$$P(X = 2) = \frac{e^{-3}3^2}{2!}$$
$$P(X = 2) = \frac{e^{-3}(9)}{2}$$
$$P(X = 2) = 4.5e^{-3}$$

Since:

$$e^{-3} \approx 0.049787$$

$$P(X = 2) \approx 0.2240$$

Answer

22.40%

Interpretation

There is approximately a **22.4% probability of exactly 2 model errors in the next 1,000 predictions.**

Problem 6: Website Clicks

An advertisement receives an average of **6 clicks per minute.**

Find the probability of receiving **no clicks** in a particular minute.

Solution

Given:

$$\lambda = 6$$

We need:

$$P(X = 0)$$

Using:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Therefore:

$$P(X = 0) = \frac{e^{-6} 6^0}{0!}$$

Since:

$$6^0 = 1$$

and:

$$0! = 1$$

we get:

$$P(X = 0) = e^{-6}$$
$$P(X = 0) \approx 0.00248$$

Answer

$$0.248\%$$

Interpretation

There is only about a **0.248% probability of receiving zero clicks in a minute** when the average is 6 clicks per minute.

Problem 7: At Least One Server Failure

A data center experiences an average of **2 server failures per month**.

Find the probability of **at least one server failure in a month**.

Solution

We need:

$$P(X \geq 1)$$

Use the complement:

$$P(X \geq 1) = 1 - P(X = 0)$$

Given:

$$\lambda = 2$$

Therefore:

$$P(X = 0) = e^{-2}$$
$$\approx 0.1353$$

Hence:

$$P(X \geq 1) = 1 - 0.1353$$
$$P(X \geq 1) \approx 0.8647$$

Answer

86.47%

Data Science Interpretation

There is approximately an **86.47% probability of at least one server failure during a month** under the assumed Poisson model.

Problem 8: No More Than 2 Customer Complaints

A company receives an average of **3 complaints per day**.

Find the probability that it receives **no more than 2 complaints** on a particular day.

Meaning

"No more than 2" means:

$$X \leq 2$$

Therefore:

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

Step 1: $P(X = 0)$

$$\begin{aligned} P(X = 0) &= e^{-3} \\ &\approx 0.0498 \end{aligned}$$

Step 2: $P(X = 1)$

$$\begin{aligned} P(X = 1) &= \frac{e^{-3} 3^1}{1!} \\ &\approx 0.1494 \end{aligned}$$

Step 3: $P(X = 2)$

$$\begin{aligned} P(X = 2) &= \frac{e^{-3} 3^2}{2!} \\ &\approx 0.2240 \end{aligned}$$

Therefore:

$$P(X \leq 2) = 0.0498 + 0.1494 + 0.2240$$

$$P(X \leq 2) \approx 0.4232$$

Answer

$$42.32\%$$

Problem 9: More Than 5 Website Requests

A small API receives an average of **4 requests per second**.

Find the probability that it receives **more than 5 requests in one second**.

Solution

We need:

$$P(X > 5)$$

Use the complement:

$$P(X > 5) = 1 - P(X \leq 5)$$

Therefore:

$$P(X > 5) = 1 - [P(X = 0) + P(X = 1) + \dots + P(X = 5)]$$

For:

$$\lambda = 4$$

the probabilities are approximately:

$$P(X = 0) = 0.0183$$

$$P(X = 1) = 0.0733$$

$$P(X = 2) = 0.1465$$

$$P(X = 3) = 0.1954$$

$$P(X = 4) = 0.1954$$

$$P(X = 5) = 0.1563$$

Therefore:

$$P(X \leq 5) \approx 0.7851$$

Hence:

$$P(X > 5) = 1 - 0.7851$$

$$P(X > 5) \approx 0.2149$$

Answer

$$21.49\%$$

Interpretation

There is approximately a **21.49% probability that more than 5 requests arrive in one second.**

This is useful when designing **API and server capacity.**

Problem 10: Mean, Variance and Standard Deviation

A website receives an average of **12 visitors per minute.**

Find:

1. Mean
2. Variance
3. Standard deviation

Solution

For Poisson distribution:

$$\lambda = 12$$

Mean

$$E(X) = \lambda$$

$$E(X) = 12$$

Variance

$$Var(X) = \lambda$$

$$Var(X) = 12$$

Standard deviation

$$\sigma = \sqrt{\lambda}$$

$$\sigma = \sqrt{12}$$

$$\sigma \approx 3.464$$

Answer

Measure	Value
Mean	12
Variance	12
Standard deviation	3.464

Problem 11: Data Preprocessing Errors

A data pipeline contains an average of **2 missing-value errors per dataset batch**.

Find the probability that a randomly selected batch contains **exactly 4 errors**.

Solution

Given:

$$\lambda = 2$$

$$x = 4$$

Therefore:

$$P(X = 4) = \frac{e^{-2} 2^4}{4!}$$

$$P(X = 4) = \frac{e^{-2} (16)}{24}$$

$$P(X = 4) = \frac{2}{3} e^{-2}$$

$$P(X = 4) \approx 0.0902$$

Answer

$$9.02\%$$

Interpretation

There is approximately a **9.02% probability** that a dataset batch contains **exactly 4 missing-value errors**.

Problem 12: Call Center Analytics

A customer-support center receives an average of **8 calls per hour**.

Find the probability that exactly **10 calls** arrive in one hour.

Solution

Given:

$$\begin{aligned}\lambda &= 8 \\ x &= 10\end{aligned}$$

Therefore:

$$P(X = 10) = \frac{e^{-8} 8^{10}}{10!}$$

Since:

$$10! = 3628800$$

we obtain:

$$P(X = 10) \approx 0.0993$$

Answer

9.93%

Data Science Interpretation

There is approximately a **9.93% probability of receiving exactly 10 calls in an hour**.

This is useful for **workforce planning and call-center resource allocation**.

Problem 13: Scaling the Time Interval

A website receives an average of **6 visitors per minute**.

What is the expected number of visitors in **5 minutes**?

Solution

Average rate:

6 visitors/minute

Time:

5 minutes

Therefore:

$$\lambda = 6 \times 5$$

$$\lambda = 30$$

Answer

The expected number of visitors in 5 minutes is:

30

Important Data Science Concept

If the rate is constant, the Poisson parameter scales with the interval:

$$\lambda_{\text{new}} = \lambda_{\text{rate}} \times \text{new interval}$$

For example:

6 per minute

becomes:

$$6 \times 60 = 360 \text{ per hour}$$

Problem 14: Fraud Detection — At Least 2 Frauds

A bank observes an average of **3 fraudulent transactions per day**.

Find the probability of **at least 2 fraudulent transactions** in a day.

Solution

We need:

$$P(X \geq 2)$$

Use the complement:

$$P(X \geq 2) = 1 - P(X < 2)$$

Therefore:

$$P(X \geq 2) = 1 - [P(X = 0) + P(X = 1)]$$

Given:

$$\lambda = 3$$

Calculate $P(X = 0)$

$$P(X = 0) = e^{-3} \approx 0.0498$$

Calculate $P(X = 1)$

$$P(X = 1) = 3e^{-3} \approx 0.1494$$

Therefore:

$$\begin{aligned} P(X \geq 2) &= 1 - (0.0498 + 0.1494) \\ P(X \geq 2) &= 1 - 0.1992 \end{aligned}$$

$$P(X \geq 2) \approx 0.8008$$

Answer

80.08%

Interpretation

There is approximately an **80.08% probability of observing at least 2 fraudulent transactions in a day.**

Problem 15: Real Data Science Problem — Server Monitoring

A cloud application receives an average of **20 requests per second**.

The data science team wants to determine the probability that exactly **25 requests** arrive during one second.

They also want to know the mean, variance, and standard deviation of the number of requests.

Solution

Given:

$$\begin{aligned}\lambda &= 20 \\ x &= 25\end{aligned}$$

Step 1: Probability

Using:

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

we get:

$$P(X = 25) = \frac{e^{-20} 20^{25}}{25!}$$

Numerically:

$$P(X = 25) \approx 0.0446$$

Therefore:

$$P(X = 25) \approx 4.46\%$$

Step 2: Mean

$$E(X) = \lambda$$

$$E(X) = 20$$

Step 3: Variance

$$\text{Var}(X) = \lambda$$

$$\text{Var}(X) = 20$$

Step 4: Standard Deviation

$$\sigma = \sqrt{\lambda}$$

$$\sigma = \sqrt{20}$$
$$\sigma \approx 4.472$$

Final Answer

Measure	Value
Average requests	20
Probability of exactly 25	4.46%
Mean	20
Variance	20
Standard deviation	4.472

Data Science Interpretation

This analysis can help a cloud engineering team understand the **natural variation in request traffic** and decide whether server capacity is sufficient.

Important Poisson Distribution Formulas

If:

$$X \sim \text{Poisson}(\lambda)$$

then:

Probability of exactly x events

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

Probability of zero events

$$P(X = 0) = e^{-\lambda}$$

Probability of at least one event

$$P(X \geq 1) = 1 - e^{-\lambda}$$

Mean

$$E(X) = \lambda$$

Variance

$$Var(X) = \lambda$$

Standard deviation

$$\sigma = \sqrt{\lambda}$$



Binomial vs Poisson vs Geometric

This comparison is **very important for Data Science students**.

Distribution	Main Question	Example
Bernoulli	What happens in one trial?	Does one customer purchase?
Binomial	How many successes in n trials?	How many of 100 customers purchase?
Geometric	How many trials until first success?	How many customers until first purchase?
Poisson	How many events occur in an interval?	How many website visitors arrive per minute?

Remember it this way:

Bernoulli → ONE trial
Binomial → SUCCESS COUNT in fixed trials
Geometric → WAIT for FIRST SUCCESS
Poisson → EVENT COUNT in a fixed interval



Poisson Distribution in Data Science

1. Web Analytics

Number of visitors per minute:

X = website visitors/minute

2. API Monitoring

Number of API requests per second:

$$X = \text{requests/second}$$

3. Fraud Analytics

Number of fraudulent transactions per day:

$$X = \text{frauds/day}$$

4. Customer Analytics

Number of complaints per day:

$$X = \text{complaints/day}$$

5. Machine Learning

Number of model errors per 1,000 predictions:

$$X = \text{errors/1000 predictions}$$

6. Data Quality

Number of missing or corrupted records in a batch:

$$X = \text{errors/batch}$$

7. Cloud Computing

Number of server failures per month:

$$X = \text{failures/month}$$

8. NLP / Text Analytics

Number of occurrences of a particular event or token in a defined collection or document window can sometimes be modeled with Poisson methods, subject to the modeling assumptions.



Key Intuition

The easiest way for students to remember Poisson is:

"I know the average rate, and I want to know how many events will occur in a fixed interval."

For example:

Average = 5 visitors/minute

Question:

What is the probability of exactly 3 visitors in one minute?

That's **Poisson**.

$$\lambda = 5, \quad P(X = 3) = \frac{e^{-5} 5^3}{3!}$$

And the beautiful property to remember is:

Poisson Mean=Poisson Variance= λ